angladesh Army University of Science and Technology (BAUST), Saidpur

Department of Computer Science and Engineering (CSE)

Class Test 1, Winter 2026

Program: **B.Sc. Engg. in CSE** Course Code: **CSE 4245 (Syl. 2021)** Credit: **3.0** Level-4, Term- II

Time: **30 (Thirty Minutes)** Course Title: **Digital Image Processing** Full Marks: **60**

Student ID:	**Name:**	**Obtained marks**	

Answer the following questions:

	Questions	**CLO**	**Marks**
Q1.	Describe the purpose of quantization in image processing and the role of the Max–Lloyd algorithm in minimizing distortion.	CLO1	10
Q2.	Briefly Explain necessity of quantization process in image processing.	CLO1	10
Q3.	Describe the necessity of correlation and convolution in image processing.	CLO2	10
Q4.	Explain necessity of histogram equalization and histogram matching in image processing.	CLO2	10
Q5.	Discuss about image segmentation process and its applications.	CLO2	10
Q6.	Explain the necessity of Image Morphological Processing.	CLO2	10

Answers:

Q1.	**Describe the purpose of quantization in image processing and the role of the Max–Lloyd algorithm in minimizing distortion.**	**CLO1**	**10**

Q2.	**Briefly Explain necessity of quantization in image processing.**	**CLO1**	**10**
Q3.	**Describe the necessity of correlation and convolution in image processing.**	**CLO2**	**10**

Q4.	**Explain necessity of histogram equalization and histogram matching in image processing.**	**CLO2**	**10**

Q5.	**Discuss about image segmentation process and its applications.**	**CLO2**	**10**

Q6.	**Explain the necessity of Image Morphological Processing.**	**CLO2**	**10**